

Umberto Zerbinati

Curriculum Vitae

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Scientific Biography

I am a DPhil student in Applied Mathematics at the University of Oxford, under the supervision of Prof. Patrick E. Farrell, and a Stipendiary Lecturer in Applied Mathematics at Oriel College, Oxford. My research interests lie in Partial Differential Equations, Numerical Analysis, and Scientific Computing. I focus on the derivation and analysis of models for fluids with ordering, such as liquid crystals, starting from a kinetic description, as well as on the design of structure-preserving finite element discretisations for the resulting equations.

Education

- October 2022 **DPhil in Mathematics, University of Oxford, Oxford, United Kingdom**
present **Supervisor:** Prof. Patrick E. Farrell.
- August 2020 **Master's Degree in Applied Mathematics, KAUST, Thuwal, Saudi Arabia**
- September 2022 **Thesis Title:** A Priori Error Analysis For A Penalty Finite Element Method.
Supervisor: Prof. Daniele Boffi. GPA: 3.74/4.
- September 2016 **Bachelor's Degree in Mathematics, University of Pavia, Pavia, Italy**
- February 2020 **Thesis Title:** Second Order Finite Difference Methods For The Wave Equation With Dirichlet Boundary Conditions.
Supervisors: Prof. Andrea Moiola and Prof. Ilaria Perugia. Graduation Grade: 106/110.
- February 2019 **Erasmus+ Traineeship, University of Vienna, Vienna, Austria**
- September 2019 **Research Topic:** Numerical solution of the wave equation.
- September 2016 **Collegio Ghislieri, Pavia, Italy**
- September 2019

Teaching Experience

- January 2025 **Stipendiary Lecturer in Applied Mathematics, Oriel College, University of Oxford, Oxford, United Kingdom**
present
- May 2025 **Visiting Lecturer on Spectral Theory and Spectral Practice, University of Edinburgh, Edinburgh, United Kingdom**
Invited by: Prof. Kaibo Hu.
- October 2023 **TA for Applied Partial Differential Equations (B5.2), Mathematical Institute, Oxford, United Kingdom**
December 2023 **Lecturer:** Prof. Andreas Muench.
- October 2023 **Tutor for Metric Spaces and Complex Analysis (A2), Wadham College, Oxford, United Kingdom**
December 2023 **Lecturer:** Prof. Yuji Nakatsukasa.
- January 2023 **Tutor for Numerical Analysis (A7), Magdalen College, Oxford, United Kingdom**
March 2023 **Lecturer:** Prof. Andrew Wathen.
- October 2022 **TA for Numerical Linear Algebra (C6.1), Mathematical Institute, Oxford, United Kingdom**
December 2022 **Lecturer:** Prof. Yuji Nakatsukasa.

Research Visits

- March 2023 **University of Catania**, *Working with Prof. Giovanni Russo*
Research Topic: Particle pushers for non-Hamiltonian systems.
- December 2019 **University of Vienna**, *Working with Prof. Anastasia Molchanova*
Research Topic: Finite element discretisations for elasticity.

Work Experience

- September 2020 **Founder and CEO of ZOEEN S.r.l.**, *Pavia, Italy*
present **ZOEEN** is a startup company developing high-power energy devices for medical applications.

Professional Memberships

- European Mathematical Society (EMS).
Society for Industrial and Applied Mathematics (SIAM).

Software Development

	Level	Skill	Comment
Language:	■■■■■	Python	<i>I contribute to software libraries PETSc and Firedrake. I develop and maintain my own software library, ngsPETSC.</i>
	■■■ ■ ■	C++	<i>I contribute to the software library Netgen and NGSolve.</i>

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[2] Clarissa Astuto, Armando Coco, and Umberto Zerbinati. A comparison of the Coco–Russo scheme and $\mathbb{P}_1$ -FEM for elliptic equations in arbitrary domains. In *ICIAM 2023*. Springer Nature Singapore, 2024.

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[4] Pablo Brubeck, Charles Parker, and Umberto Zerbinati. Achieving material robustness via symmetric stress finite element discretizations, 2026. <https://arxiv.org/abs/2605.21425>.

[5] Jose A. Carrillo, Patrick E. Farrell, Andrea Medaglia, and Umberto Zerbinati. A kinetic theory approach to ordered fluids, 2025. <https://arxiv.org/abs/2508.10744>.

[6] Patrick E. Farrell, Josef Málek, Ondřej Souček, and Umberto Zerbinati. A kinetic interpretation of thermomechanical restrictions of continua. *International Journal of Engineering Sciences*, 2026.

[7] Patrick E. Farrell, Giovanni Russo, and Umberto Zerbinati. Kinetic derivation of an inviscid compressible leslie–ericksen equation for rarified calamitic gases. *Multiscale Modeling & Simulation*, 22(4):1585–1607, 2024.

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- [10] Lorenzo Lazzarino, Yuji Nakatsukasa, and Umberto Zerbinati. Preconditioned normal equations for solving discretised partial differential equations, 2025. <https://arxiv.org/abs/2502.17626>.
- [11] Manuel Trezzi and Umberto Zerbinati. When rational functions meet virtual elements: the lightning virtual element method. *Calcolo*, 61(3):35, Jun 2024.
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- [14] Tim van Beeck and Umberto Zerbinati. An adaptive mesh refinement strategy to ensure quasi-optimality of the conforming finite element method for the helmholtz equation via t-coercivity. *arXiv preprint (2403.06266)*, 2024.
- [15] Stefano Zampini, Umberto Zerbinati, George Turkyiah, and David Keyes. Petscml: Second-order solvers for training regression problems in scientific machine learning. In *Proceedings of the Platform for Advanced Scientific Computing Conference, PASC '24*, New York, NY, USA, 2024. Association for Computing Machinery.
- [16] U. Zerbinati. PINNs and GaLS: A priori error estimates for shallow physics informed neural networks applied to elliptic problems. *IFAC-PapersOnLine*, 55(20):61–66, 2022. 10th Vienna International Conference on Mathematical Modelling MATHMOD 2022.